Osteoporosis Risk Prediction Using Machine Learning

A Project Report

In the partial fulfillment of the award of the degree of

B. Tech

at

Ardent Computech Pvt. Ltd

Submitted by

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Swami Vivekananda University

Certificate from the Mentor

This is to certify that **Sukalpa Manna, Sudipta Giri** has completed the project titled

Osteoporosis Risk Prediction Using Machine Learning

under my supervision during the period from **April** to **May** which is in partial fulfillment of requirements for the award of the **B.Tech** and submitted to Department **Computer Science & Engineering** of **Swami Vivekananda University**

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Signature of the Mentor

Date:

Acknowledgment

I take this opportunity to express my deep gratitude and sincerest thanks to my project mentor, **Angira Banarjee** for giving the most valuable suggestions, helpful guidance, and encouragement in the execution of this project work.

I would like to give a special mention to my colleagues. Last but not least I am grateful to all the faculty members of the **Ardent Computech Pvt. Ltd.** for their support.

Abstract

The increasing prevalence of osteoporosis has become a major global healthcare concern, particularly among elderly individuals and postmenopausal women. Osteoporosis is a medical condition characterized by reduced bone density and deterioration of bone tissue, leading to an increased risk of fractures and other severe complications. Early prediction and identification of high-risk individuals can significantly improve prevention strategies and treatment outcomes.

This project presents a comprehensive machine learning-based approach for Osteoporosis Risk Prediction using supervised learning algorithms. The primary objective of this system is to analyze patient medical and lifestyle-related information and predict whether a person is at risk of developing osteoporosis.

The dataset used in this project contains multiple health-related attributes such as Age, Gender, Hormone levels, Family History, Race, Weight, Calcium Intake, Physical Activity, Smoking habits, Medical Conditions, Medications, and Fracture history. These attributes are processed and analyzed to uncover hidden patterns and relationships associated with osteoporosis.

The implementation involves various stages including data preprocessing, handling missing values, encoding categorical variables, feature scaling, model training, testing, and performance evaluation. Three machine learning classification algorithms were implemented and compared: Logistic Regression, Random Forest Classifier, and Support Vector Machine (SVM).

The performance of each model was evaluated using accuracy score and confusion matrix. Among all implemented models, the Support Vector Machine (SVM) achieved the highest accuracy of 83.42%, making it the most effective model for this prediction task.

Additionally, the project includes a risk profiling system that categorizes individuals into Low Risk, Medium Risk, and High Risk groups based on probability outputs generated by the best-performing model. This can assist healthcare professionals in making timely and informed medical decisions.

Overall, this project demonstrates how machine learning techniques can be effectively applied in the healthcare sector for disease prediction, early diagnosis, and preventive treatment planning.

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Introduction

Osteoporosis is a serious bone disease that reduces bone density and increases the likelihood of fractures. It affects millions of people worldwide, especially older adults and women after menopause. Early detection of osteoporosis risk can help prevent severe medical complications and improve quality of life.

Machine learning has become a powerful tool in healthcare for analyzing medical datasets and predicting diseases based on patient history and health indicators. By training models on historical data, healthcare professionals can make faster and more accurate decisions.

This project focuses on building a machine learning-based system to predict osteoporosis risk using patient lifestyle and medical information.

Problem Statement

The main objective of this project is to develop a predictive model that can classify whether a person is at risk of osteoporosis based on medical and lifestyle-related features.

Specific challenges include:

1. Handling missing values in the dataset.
2. Encoding categorical data into numerical form.
3. Selecting the best classification algorithm.
4. Evaluating model performance accurately.
5. Profiling individuals into risk categories.

Objectives

1. Load and understand the dataset structure.
2. Identify input and target variables.
3. Clean the dataset by handling missing values.
4. Encode categorical variables.
5. Split the dataset into training and testing sets.
6. Apply feature scaling.
7. Train multiple classification algorithms.
8. Compare model accuracies.
9. Generate confusion matrices.
10. Perform risk profiling.

Scope

The scope of this project is limited to predicting osteoporosis risk using the provided dataset. The system analyses patient-related health information and predicts whether the patient is at risk.

This project includes:

• Data preprocessing

• Feature engineering

• Machine learning model building

• Performance evaluation

• Risk profiling

This project does not include:

• Real-time hospital integration

• Live patient monitoring

• Clinical diagnosis replacement

Technology Stack

Python	**Programming Language**
Jupyter Notebook	**Development Environment**
Pandas	**Data Manipulation**
NumPy	**Numerical Computation**
Matplotlib	**Visualization**
Seaborn	**Data Visualization**
Scikit-learn	**Machine Learning Models**

System Architecture

The system follows these steps:

1. Data Collection
2. Data Preprocessing
3. Feature Encoding

4. Train-Test Split
5. Feature Scaling
6. Model Training
7. Prediction
8. Risk Profiling
9. Result Visualization

Dataset Details

The dataset contains the following columns:

Id, Age, Gender, Hormone, FHistory, Race, Weight, CalciumIn, Activity, Smoking, MedCondition, Medications, Fractures, Osteoporosis.

Target Variable:

Osteoporosis: 0 = No Risk

1 = Risk

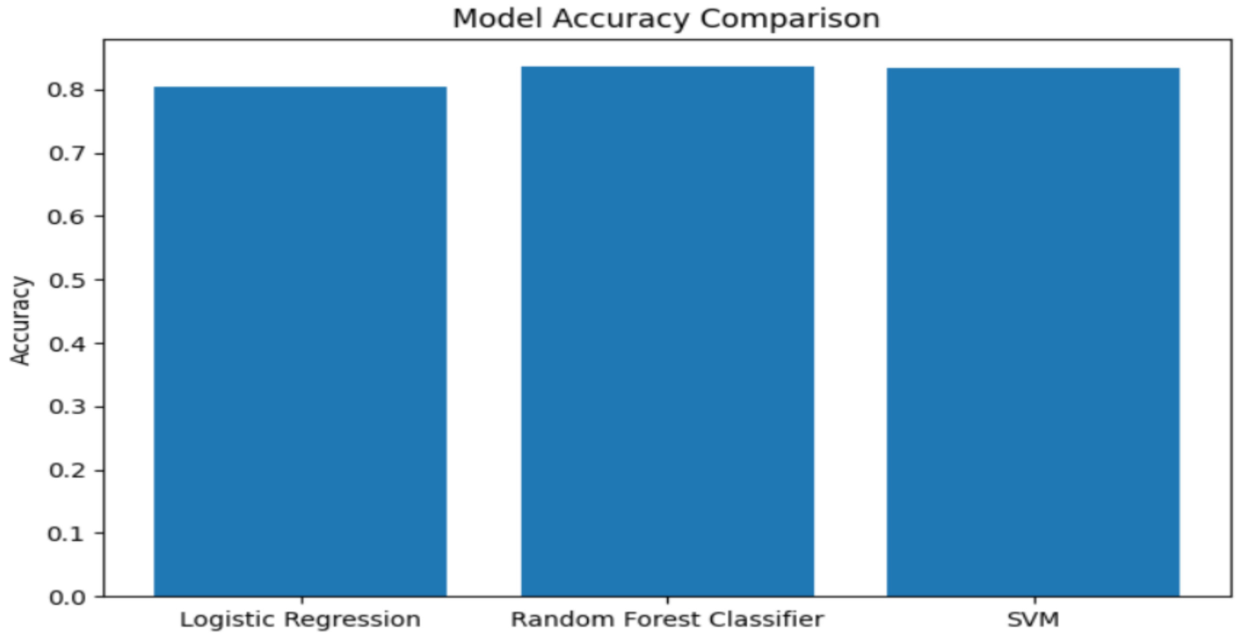
Implementation

The implementation includes:

- Loading dataset
- Checking null values
- Filling missing values using mean and mode
- Encoding categorical values using LabelEncoder
- Splitting data
- Scaling features
- Training Logistic Regression, Random Forest, and SVM
- Predicting and evaluating accuracy

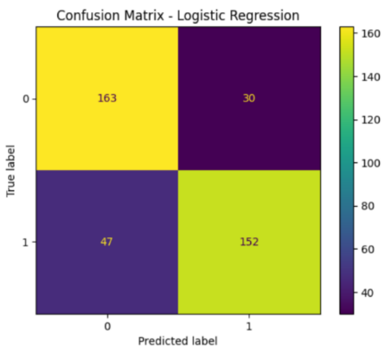
Model Performance

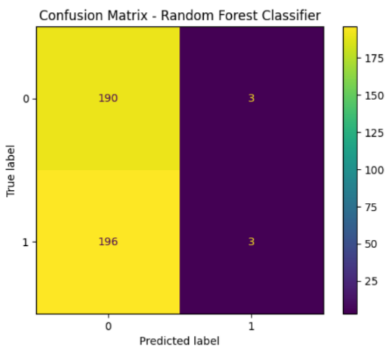
Model	Accuracy
Logistic Regression	80.36%
Random Forest	83.67%
SVM	83.42%

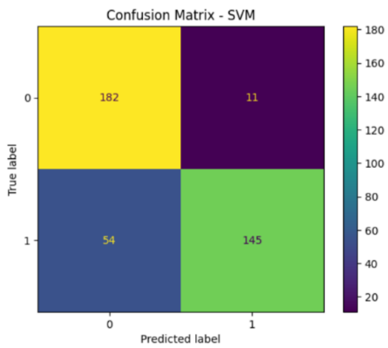


Confusion Matrix

Confusion matrices were generated for all three models to evaluate True Positive, True Negative, False Positive, and False Negative values.







Risk Profiling

Individuals are categorized as:

- Low Risk → Probability < 0.33
- Medium Risk → Probability between 0.33 and 0.66
- High Risk → Probability > 0.66

```python
best_model_name = max(accuracies, key=accuracies.get)
best_model = models[best_model_name]

print("Best Model:", best_model_name)

if best_model_name == "Random Forest":
    probs = best_model.predict_proba(x_test)[:,1]
else:
    probs = best_model.predict_proba(x_test_scaled)[:,1]

risk_profile = []

for p in probs:
    if p < 0.33:
        risk_profile.append("Low Risk")
    elif p < 0.66:
        risk_profile.append("Medium Risk")
    else:
        risk_profile.append("High Risk")


result = pd.DataFrame({
    "Actual": y_test.values,
    "Probability": probs,
    "Risk Level": risk_profile
})

print(result.head(10))
```

```
Best Model: Random Forest Classifier
   Actual  Probability   Risk Level
0       0         0.36  Medium Risk
1       1         0.27     Low Risk
2       1         0.37  Medium Risk
3       1         0.20     Low Risk
4       0         0.36  Medium Risk
5       1         0.25     Low Risk
6       1         0.30     Low Risk
7       0         0.21     Low Risk
8       1         0.41  Medium Risk
9       1         0.28     Low Risk
```

Testing

Testing was performed by:

- Checking missing values
- Validating data types
- Evaluating model accuracy
- Verifying prediction outputs

Challenges

- Missing values in medical columns
- Handling categorical data
- Selecting best-performing algorithm
- Balancing model accuracy and speed

Future Enhancements

- Hyperparameter tuning
- Deep learning implementation
- Web deployment
- Real-time medical data integration

Conclusion

This project successfully predicts osteoporosis risk using machine learning techniques. Among the tested models, Support Vector Machine performed best with 83.42% accuracy.

The project demonstrates the usefulness of machine learning in healthcare applications.